Margaret E. B. Webb

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Education

Ph.D. Engineering Education May 2025 (4.0 GPA) **Virginia Tech**, Blacksburg, VA

- *Advisor*: Dr. Marie C. Paretti
- *Dissertation:* Exploring Systems Influences on Engineering Graduate Student Interdisciplinary Identity-Based Motivation towards Research Careers
    - Nominated for Council of Graduate Studies (CGS) / ProQuest Distinguished Dissertation Award by Virginia Tech Graduate School
    - VT Department of Engineering Bevlee Watford Outstanding Dissertation
    - Nominated for VT Graduate School Outstanding Dissertation for Social Sciences
    - 1st place VT COE Torgersen Outstanding Dissertation Award

M.S. Civil Engineering May 2024 (3.95 GPA) **Virginia Tech**, Blacksburg, VA

- *Focus Area:* Sustainable Land Development

Graduate Certificate: Disaster Resilience and Risk Management (4.0 GPA) **Virginia Tech**

B.S. Mechanical Engineering: Magna Cum Laude (3.88 GPA) **Rice University,** Houston, TX

Research Work Experience

Cornell University, Office of Inclusive Excellence (OIE) College of Engineering Ithaca, NY
Teaching and Learning Consultant, Engineering Teaching Assistant and Course Assistant Professional Development Program 5/2026 – Present

- *Supervisors:* Dr. Celia Evans and Dr. Wenjing Luo
- Co-developing and co-facilitating evidence-based, interactive training workshops for engineering graduate Teaching Assistants (TAs) and undergraduate Course Assistants (CAs).
- Co-presenting and supporting logistics for the two-day graduate TA Development program and the Masters/Undergraduate CA Training event each semester.
- Conducting microteaching sessions with small groups of TAs during the first two weeks of each semester to support instructional growth and reflective practice.
- Participating in a structured professional development series focused on student-centered pedagogy, workshop design, and facilitation skills.
- Conducting research on the TA/CA Development Program.

Cornell University, Office of Inclusive Excellence (OIE) College of Engineering Ithaca, NY
Radical Humanity in Research (RHR) Program, Facilitator and Researcher 9/2025 – Present

- *Supervisors:* Dr. Laura Schoenle and Dr. Alex Werth and Dr. Alex Coso Strong
- Supporting a team-based professional development program for faculty and their research

groups focused on cultivating values-centered, inclusive research cultures; involves facilitated dialogues engaging faculty, graduate students, postdoctoral associates, and undergraduate researchers in co-constructing shared research group values, mutual expectations, and feedback and accountability practices.

- Conducting research on the program's impact.
- Contributing to grant writing to support program development and expansion.

Cornell University, Department of Biomedical Engineering (BME) Ithaca, NY
Postdoctoral Fellow, Cornell Disciplinary-Based Education Research (CDER) 7/2025-Present

- *Supervisors:* Dr. Alex Werth and Dr. Alex Coso Strong
- Conducting independent and collaborative research on:
    - Postdoctoral professional development (PD): exploring how structures, systems, cultures, impact postdoctoral associates as scholars and change agents.
    - Generative Artificial Intelligence's role in automating qualitative assessment of student reflection quality contextualized to specific STEM Education courses.
    - Active Learning Initiatives, programs seeking to implement evidence-based teaching strategies to innovate in education based on research, as instructional change models
- Supporting research grant writing.
- Initiating, planning, and implementing a regional Disciplinary-Based Education Research (DBER) conference.
    - Built and maintaining NYDBER conference website: https://nydber2026.github.io/nydberconference2026/
- Mentoring graduate students and undergraduate students in conducting STEM Education Research.

Virginia Tech, Office of the Dean of the College of Engineering Blacksburg, VA
Graduate Research Assistant for Graduate & Professional Studies June 2024 – May 2025
Supervisor: Dr. Holly Matusovich

- Coordinated and hosted Virginia Tech Dissertation Institute (VTDI) for graduate student writing and research skills support.
- Organized College of Engineering Graduate Student Fall Welcome Fair for student orientation and networking.
- Developed and managed 2024-2025 workshop series for professional development, targeting graduate students and external fellowship holders.
- Evaluated programming effectiveness and prepared formative and summative assessment reports for the Dean.
- Initiated and developed a college-level Canvas Site as a repository for engineering graduate student professional development materials and programming.
- Developed a peer-mentoring program for external fellowship-holding students, piloting a GPS Student Ambassador Program.
- Served on the 2024 search committee for a new Assistant Director of Graduate and Professional Studies, contributing to position description writing, application screening, interviewing, and evaluation of candidates and the COE search process.
- Organized hire for a computer scientist graduate assistant.
- Collaborated in developing a scalable and efficient computerized process for addressing COE under-enrolled classes for semesterly audits and reporting.

Virginia Tech, *Institute for Creativity, Arts, and Design* Blacksburg, VA
Designing Technology for Hospital Education, Graduate Research Assistant 2024- 2025

- Supported VT Institute for Creativity, Arts, and Technology (ICAT), Virginia Tech Major SEAD Research Grant ($40,000) *PIs: Dr. Finda Williams and Dr. Sweta Baniya*
- *Research Data Collection and Analysis:* Conducted semi-structured interviews with hospital educators, developed a qualitative codebook, and analyzed qualitative journal and interview data using qualitative research methods.
- *Research Writing*: Published and writing conference papers and journal articles related to this research.

Virginia Tech, *Engineering Education* Blacksburg, VA
Disaster Resilience and Risk Management, Graduate Research Assistant 2021 –2024

- Supported NSF Award # 1735139: Research Traineeship Program: Creating quantitative decision-making frameworks for multi-dimensional and multi-scale analysis of hazard impact. *Supervisor: Dr. Marie Paretti; PI: Dr. Robert Weiss*
- *Program Development and Assessment:* Supported the design of and conducted program-level and course-level assessments (including pre/post concept maps and annual interviews with student participants); summarized assessment findings for the annual NSF NRT report and Virginia Tech, presented findings to the DRRM advisory board and the graduate school.
- *Research Data Collection and Analysis:* Conducted semi-structured interviews and focus groups with participants, developed a qualitative codebook, and analyzed qualitative journal and interview data using qualitative research methods.
- *Research Writing:* Published conference papers and journal articles related to this research.
- *Conference Planning and Coordination:* Planned and coordinated the graduate student programming for the annual NRT (NSF Research Traineeship) conference for 2022-2023.
- Mentored junior Engineering Education graduate students in research methods, paper, and dissertation ideation, completion of PhD milestones, and career development, including navigating the job market.

Virginia Tech (VT), Center for Coastal Studies Blacksburg, VA
COVID-19 & Climate Change: Mapping the effects of compounding Crisis in the Coastal Zone using Natural Language Processing, Graduate Research Assistant 2021 – 2024

PIs: Dr. Andrew Katz and Dr. Sweta Baniya
Center for Coastal Studies, Virginia Tech Major SEAD Research Grant

- Supported development of an end-to-end method leveraging GPT-3.5 for generating qualitative codebooks from disaster-related social media data with minimal human input.
- Contributed to interdisciplinary research using NLP for qualitative analysis, demonstrating the potential of transformer-based large language models and generative AI for scaling up and expediting qualitative analysis of social media conversations around compounding crises.
- Mentored undergraduate students from Urban Planning, Geography, Communications, and Computer Science in applying natural language processing to interdisciplinary research on disaster resilience, guiding them in research writing and exploring technological solutions for compounding crises like COVID-19 and climate change.

VT, Center for Refugee, Migrant, and Displacement Studies Blacksburg, VA
Towards Digital Justice: Developing a US Citizenship App & Website with Refugees & Immigrants, GRA 2021 – 2024

PIs: Dr. Sweta Baniya and Dr. Katrina Powell
American Council of Learned Societies (ACLS) Digital Justice Seed Grant

- Supported a community-engaged, participatory action research project aimed at reimagining the U.S. naturalization process for people seeking refuge through a lens of digital justice, meaningful integration, and centering community expertise.
- Employed theoretical frameworks of knowledge justice, digital justice, and epistemic integration to critically examine the unintended consequences and structural barriers created by the integration of information and communication technologies (ICTs) into the naturalization process.
- Investigated the normative assumptions about migrants and integration embedded in digitized naturalization processes and resources, and explored the knowledge, experiences, and counter-expertise of people seeking refuge that are not represented in current system designs.
- Adopted a qualitative, community-based approach to mitigate the systemic negative effects of ICT implementation in the naturalization process and co-designed ICTs with newcomers (informed by the lived experiences and values of refugee and immigrant communities).
- Mentored undergraduate students across Computer Science, Psychology, Technical Communications, and Engineering in technology development, participatory action research, and publication strategies, fostering interdisciplinary collaboration.

VT, Center for Refugee, Migrant, and Displacement Studies Blacksburg, VA
Strengthening Virginia's Ability to Provide Migration Support and Improve Refugee Population Outcomes: Collating Refugee Resources and Collaborative Initiatives at Virginia Higher Education Institutions, GRA

PIs: Dr. Sweta Baniya and Dr. Katrina Powell

4-VA Collaborative Research Grant in collaboration with JMU, GMU, and VCU

- Supported the development of a centralized, sustainable, and public database and website that builds upon JMU's CISR's Global CWD Repository, a free resource to consolidate information on Resettlement Services for New Americans in Virginia

VT, Center for Refugee, Migrant, and Displacement Studies Blacksburg, VA
Critical Displacement Studies Fellow Aug 2021 – May 2024

PIs: Dr. Katrina Powell and Dr. Rebecca Hester

National Endowment for the Humanities in partnership with the Social Science Research Council

- Collaboratively wrote an open-access reader, the Interdisciplinary and Critical Displacement Studies Reader, that brings together a breadth of perspectives across disciplines and a variety of forms, building a Critical Displacement Studies methodological research approach.

Rice University, Mechanical Engineering Houston, TX
Rice Tribomechadynamics Lab, Undergraduate Research Assistant 2016 - 2019

- Designed and fabricated fixtures for Dr. Brakes Tribomechadynamics lab, including tensile testing tools, trunnions, and drop testing equipment
- Conducted surface dimpling tests and data analysis in support of a PhD students' dissertation
- Coordinated the onboarding of new undergraduate research assistants into a research environment

Teaching Experience

Teaching and Learning Consultant, Engineering TA/CA Development Program
Cornell University, Office of Inclusive Excellence Ithaca, NY 05/2026-Present

- Co-developed and co-facilitated evidence-based training workshops for engineering graduate TAs and undergraduate CAs, supporting the development of student-centered instructional practices across Cornell Engineering.
- Conducted microteaching sessions with small groups of TAs to provide structured, reflective feedback on teaching practice during the first weeks of fall and spring semesters.

Graduate Teaching Assistant Intercultural Issues in Professional Writing ENGL 3834
Virginia Tech, Blacksburg, VA Spring 2024, Fall 2025

- Taught modules on participatory action and community-engaged technical communication, effective presentations for conferences
- Graded weekly discussion posts and course mid-semester projects

Graduate Teaching Assistant Interdisciplinary Research Methods in Disaster Resilience GRAD 5134 2023-2024

- Contributed to course design, syllabus development, and assessment development for interdisciplinary undergraduate and graduate courses.
- Responsible for grading weekly assignments and project assignments, providing detailed feedback to 25 students.
- Collaborated with faculty instructors on iterative improvements to course materials and teaching methods based on assessment data.

Middle and High School Math Teacher
Harmony Public Schools 12/2020 – 6/2021

- Designed lesson plans with active learning pedagogies for in person, online, and
- Advanced student learning by contextualizing subject matter to student interests
- Cultivated relationships with 120 students, tens of faculty, and parents
- Collaborated with teaching teams to increase the use of active learning pedagogies in the math department

Texas Education Agency, Teaching Certification Program 5/2021

- TExES 274: Grades 6-12 Mathematics, Physical Science, and Engineering Content Exam (276/300)

Solids Mechanics Undergrad Learning Assistant, Mechanical Engineering Department
Rice University, Houston, TX Fall 2017

- Responsible for supporting recitation sessions.
- Collaborated with faculty instructors on improvements to lab materials.

Publications

Peer-Reviewed Conference Papers

1. Deters, J. R., Menon, M., Paretti, M. C., & **Webb, M.E.B.** (2021). Imagining the future interdisciplinary scholar: Exploring interdisciplinary identity development using possible selves. *Proceedings of the joint Research in Engineering Education Symposium (REES) & Australasian Association for Engineering Education Conference, Perth, Australia* [virtual].
2. **Webb, M.E.B.,** Deters, J.R., Menon, M., & Paretti, M.C. (2022). Building a Sustainable University-Wide Interdisciplinary Graduate Program to Address Disasters. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Milwaukee, Wisconsin*.
3. **Webb, M.E.B.** & Paretti, M.C. (2022). Exploring Professional Identity Development Research on Displaced Higher Education Students. *Proceedings of the Frontiers in Education (FIE) Conference, Uppsala, Sweden.*
4. Deters, J.R., **Webb, M.E.B.**, & Menon, M. (2022). Exploring Mechanical Engineering Student Perspectives of Academic Integrity During COVID-19. *Proceedings of the Australasian Association for Engineering Education Conference, Western Sydney University.*
5. Menon, M., **Webb, M.E.B.,** Josiam, M., & Deters, J.R. (2023) Sustainable Development Goals in Engineering Curriculum: A Document Analysis on Discourse and Suggestions for Improvement. *Proceedings of the Engineering Education for Sustainable Development (EESD) Conference, Colorado.*
6. **Webb, M.E.B.** & Paretti, M.C. (2023). Influences on Displaced Engineering Student Professional Identity Development: A Scoping Literature Review Across Forced Migration Contexts. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland.*
7. **Webb, M.E.B.** & Paretti, M.C. (2023). Investigating Graduate Students' Perspectives of Influences on Interdisciplinary Scholar Identity Development: An Ecological Systems Theory Approach. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland.*
8. Menon, M., **Webb, M.E.B.,** & Paretti, M.C. (2023). Understanding Faculty Perspectives of Interdisciplinary Graduate Programs. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland.*
9. Bose, B., Brotherton, H.B., Hopper, T., Oquendo, P.M., Wang, L.M., **M.E.B. Webb**, & Wilkinson, H. (2023). Graduate Student and Postdoctoral Fellow Perspectives on Advancing Women and Gender Equity in Engineering - for the Next 130 Years. *Proceedings of the American Society of Engineering Education (ASEE) Conference, Bethesda, Maryland.*
10. Scott, L., **Webb, M.E.B.,** Baniya, S., Powell, K., Salem, A. (2023). Towards Digital Justice: Developing US Citizenship Application & Website with Refugees & Immigrants. *Proceedings of the 2023 Conference on Community Writing in Denver, Colorado.*
11. Scott, L., **Webb, M.E.B.,** Baniya, S., Schell, E., Powell, K., Salem, A. (2024). Expanding Writing Abundance in Partnership with the Community. *Proceedings of the 2024 Conference on College Composition and Communication in Spokane, Washington: Writing Abundance: Celebrating 75 Years of Conversations about Rhetoric, Composition, Technical Communication, and Literacy.*
12. **Webb, M.E.B.**, Feng, X., Aarnio, H., Sundman, J., Bilow, F., Taka, M., Paretti, M., Keskinen, M. (2024). Finding Common Ground: Comparing Engineering and Design Graduate Students' Conceptualizations of Interdisciplinary Education across Two Institutions. *Proceedings of the 2024 American Society of Engineering Education (ASEE) Conference, Portland, Oregon.*

13. **Webb, M.E.B.** & Paretti, M.C. (2024). Understanding Ecosystems of Interdisciplinary Graduate Education through an Ecological Systems Approach. *Proceedings of the 2024 American Society of Engineering Education (ASEE) Conference, Portland, Oregon.* *<u>**Won ASEE 2024 Graduate Studies Division Best Student Paper Award**</u>*
14. **Webb, M.E.B.**, Baniya, S., Powell, K., El Kadima, M., Drumgold-Smith, M., & Rissler, G. (2024). Towards Digital Justice and Consolidated Resettlement Resources for New Americans: A Virginia Database Initiative. *Proceedings of the 2024 ARNOVA Conference, Washington, DC.*
15. **Webb, M.E.B.** & Paretti, M.C. (2025). Tracking the Evolution of Interdisciplinary Identity-Based Motivation in Engineering Graduate Students: A Longitudinal Case Study. *Proceedings of the 2025 American Society of Engineering Education (ASEE) Conference, Montreal, Canada.* *<u>**Won ASEE 2025 Graduate Studies Division Best Student Paper Award**</u>*
16. **Webb, M.E.B.,** Coso Strong, A. & Werth, A. (2026). Generative Stories Generating Research Insight: GenAI-Enabled Research Fiction for Examining Multi-Perspective "Critical Tensions" in Scholarly Development. *Proceedings of the 2026 American Society of Engineering Education (ASEE) Conference St. Lawrence Division, Cornell University, Ithaca, NY.*
17. **Webb, M.E.B.,** Coso Strong, A. & Werth, A. (2026). Building Capacity for Instructional Change: A Critical Review and Conceptual Framework for Discipline-Based Education Research (DBER) Postdoctoral Professional Development. *Proceedings of the 2026 American Society of Engineering Education (ASEE) Conference, Charlotte, North Carolina.*
18. **Webb, M.E.B.,** McColley, C., Alrizqi, M. Manuel Sanders, G.., & Werth, A. (2026). Toward Scalable Assessment of Undergraduate Reflective Practice: Comparing Multiple Reflection Quality Codebook Validation Approaches. *Proceedings of the 2026 American Society of Engineering Education (ASEE) Conference, Charlotte, North Carolina.*
19. Jaramillo, J., **Webb, M.E.B.,** & Coso Strong, A. (2026). Strategies for Hybrid, Hands-on Courses in Robotics, Embedded Systems, and IoT. *Proceedings of the 2026 American Society of Engineering Education (ASEE) Conference, Charlotte, North Carolina.*
20. In Progress: Campos Oaxaca, G., Webb, M.E.B., Werth, A. (2027) Who Switches? Field Pivoting, Backgrounds, and Career outcomes in the Survey of Doctorate Recipients. *Proceedings of the 2027 AERA Conference.* Toronto, CA.

Peer-Reviewed Journal Articles

1. **Webb, M.E.B.,** Singh, H., Inman, R., Baniya, S. & Katz, A. (2025). Advancing Qualitative Analysis in Professional Disaster and Risk Communication: A Comparative Study of an OpenAI ChatGPT 3.5 Model-Enabled Method for Processing Complex Public Discourse. *International Journal of Disaster Risk Reduction.*
2. **Webb, M.E.B.** & Paretti, M. (2025). Navigating (Inter)disciplinary systems: ecological systems analysis of engineering graduate students' motivation for interdisciplinary development. *European Journal of Engineering Education Special Issue: Interdisciplinary Learning and Transformation of Engineering Education.*
3. **Webb, M.E.B.**; Baniya, S.; Smith, A.; Williams, I.F. (March 2026). U.S. Hospital Educators' Technology Needs: A Qualitative Study for Developing Action-Oriented Technology. *IEEE Transactions on Professional Communication.*

4. **Webb M.E.B**., Baniya S, Powell K, Rizvi M, Scott L, Dromgold-Sermen M, et al. (2026). Reimaging Information Architecture: Community-Centered and Collaborative Digital Infrastructure in Refugee Service Ecosystems. *Computers and Composition.*
5. Journal Article In Revision: **Webb, M.E.B.** & Paretti, M. (estimated 2026). Beyond Barriers and Supports: A Systems Interaction Analysis of How Microsystems Work Together to Shape Interdisciplinary Development in Engineering Graduate Education. *Journal of Engineering Education Special Issue on Surrounding Systems' Influences on Engineering Education: Focus on Organizations, Policies, and Contexts.*
6. Journal Article in Progress: **Webb, M.E.B.** & Paretti, M. (estimated 2026). Visualizing Interdisciplinary Scholar Development: A Longitudinal Analysis of Engineering Graduate Students' Identity-Based Motivation Trajectories for Interdisciplinary Research Careers. *Studies in Graduate and Postdoctoral Education.*
7. Journal Article in Progress: **Webb, M.E.B.,** Coso Strong, A, & Werth, A. (estimated 2026). Cycles of Recognition: How DBER Postdocs Build Professional Agency for Instructional Change with Archetypal Patterns. *Studies in Graduate and Postdoctoral Education.*
8. Journal Article in Progress: **Webb, M.E.B.,** Mendez-Sanders, G., McColley, C., Kim, S., & Werth, A. (estimated 2026). Not All Codes Scale Alike: Dataset Balancing, Code Type, and Interrater Agreement in GenAI-Enabled Qualitative Research. *Studies in Engineering Education.*
9. Journal Article in Progress: Lightner, T.L., **Webb, M.E.B.**, Schilling, M. R. (estimated 2027) Rethinking Self-Authorship Through Critical Resilience Theory: Supporting Marginalized STEM Graduate Students in Interdisciplinary Spaces. *Frontiers in Education Special Issue on From Gatekeeping to Gatebreaking: Leading Through Sociopolitical Pressures to Reimagine STEMM Ecosystems.*
10. Journal Article in Progress: **Webb, M.E.B.,** Coso Strong, A, & Werth, A. (estimated 2027). Not Just the Postdoc: Mutual Learning from Critical Tensions in Postdoc-Faculty Dyads. *International Journal of STEM Education.*
11. Journal Article in Progress: Campos Oaxaca, G. and **Webb, M.E.B**., Werth, A. (estimated 2027). Discipline switching and career outcomes among U.S.doctorate recipients: Evidence from the Survey of Doctorate Recipients, 2001—2023. *Nature.*

Book Chapters

1. Baniya, S., Powell, K., Salem, A., **Webb, M.**, Scott, L. (2025). Incorporating Community Knowledge in Design: A Reflective Account of Designing Technology with Justice. In J. Tham & J. Zhang (Eds). Designing for Social Justice: Community-engaged Approaches to Technical and Professional Communication. *Routledge ATTW Book Series.*
2. Accepted: Baniya, S., Powell, K., Salem, A., **Webb, M.**, Scott, L., Williams, C., & East, A. (Estimated 2026). Enacting Digital Justice in Partnership with Displaced Refugee and Immigrant Communities. In Powell, K, et.al., (Eds) Critical Displacement Studies Reader. *Virginia Tech Publishing.*
3. In Revisions: Feng, X., Aarnio, H., **Webb, M.E.B.,** Schiebel, J., Hingle, A. (estimated 2027). The Hidden Curriculum of a PhD: From Getting Started To Getting It Done in Engineering Education Research. *International Handbook of Engineering Education Research.*

Blog Posts

1. **Webb, M.E.B.,** Menon, M., Deters, J.R., & Paretti, M.C. (2024). Breaking Silos: Rethinking University Structures to Facilitate Interdisciplinary Engineering Graduate Education. *European Journal of Engineering Education Special Issue: Interdisciplinary Learning and Transformation of Engineering Education.*

Presentations

Invited Talks and Workshops

1. **Webb, M.E.B.** (2020, March). "Engineering Professional Development Workshop: Time Management Skills, Professional Communication, and the Job Hunt." Invited talk by Dr. Eleazar Marquez and the Mechanical Engineering Department at Rice University in Houston, TX. *30 attendees.*
2. **Webb, M.E.B.** (2022, Spring). "Engineering Work Abroad" and "Culture in Engineering." Invited talks by Dr. Jessica Deters and Dr. Homero Murzi at Virginia Tech Rising Sophomore Abroad (RSAP) Program. *40 attendees.*
3. **Webb, M.E.B.** & Singh, H. (2022, September). "Diversity Equity and Inclusion Research in Natural Hazards Engineering." Invited talk with Dr. Robin Nelson and the Natural Hazards Engineering Research Infrastructure's (NHERI) Graduate Student Council. *70 attendees.*
4. **Webb, M.E.B.** & Lightner, T. (2022, October). "Communicating Interdisciplinary Research: Taking Convergent Research Topics to Research Questions." Invited talk with Dr. Robert Weiss and Dr. Marie Paretti Virginia Tech Disaster Resilience and Risk Management Program. *20 attendees.*
5. **Webb, M.E.B.** (2022, October). "Exploring the Professional Identity Development of Displaced Engineering Students." Invited talk at Ohio State University for ENGE Graduate Student Seminar Exchange. https://eed.osu.edu/events/2022/11/seminar-series-two-one-virginia-tech-eed-phd-candidates. *30 attendees.*
6. **Webb, M.E.B.,** Singh, H, Inman, R., Katz, A., & Baniya, S. (2022, November). "COVID-19 & Climate Change: Methods for Mapping the Effects of a Compounding Crisis in the Coastal Zone" Invited talk with Dr. Robert Weiss and Virginia Tech Disaster Resilience and Risk Management Program and Center for Coastal Studies. *15 attendees.*
7. **Webb, M.E.B.** & Singh, H. (2023, February). "Cultural Competence in Hazards and Disaster Research: CONVERGE Training Module Workshop." Invited talk with Dr. Robin Nelson and the Natural Hazards Engineering Research Infrastructure's (NHERI) Graduate Student Council. https://www.youtube.com/watch?v=a4MKgKWjRjM&t=52s. *50 attendees.*
8. **Webb, M.E.B.**, Baniya, S., East, A., Williams, C., & Powell, K. (2023, February). "Towards Digital Justice: Participatory Action Research on Structural Barriers to US Citizenship for People Seeking Refuge." Invited talk with Dr. Janine Joseph at Virginia Tech Introduction to Displacement Studies Program. *25 attendees.*
9. **Webb, M.E.B.**, Baniya, S., & Powell, K. (2023, March). "Towards Digital Justice: Participatory Action Research on Structural Barriers to US Citizenship for People Seeking Refuge." Invited talk with VT Center for Refugee, Migrant, and Displacement Studies and VA CRMDS Consortium. *50 attendees.*
10. **Webb, M.E.B.**, Gerhardt, M., & Schibelius, L. (2023, Oct). "Navigating Engineering Industry: The Job Search and Beyond." Invited talk with Virginia Tech Interdisciplinary

Senior Capstone course about the transition to engineering work. *60 attendees.*

11. **Webb, M.E.B.** (2024, September). "Planning to Apply for Engineering Graduate School Fellowships." Invited talk with the VT College of Engineering Graduate and Professional Studies program. *25 attendees.*
12. **Webb, M.E.B.** (2025, December). "Conference and Poster Planning for Undergraduate Technical Communication Students." Invited talk and poster judge with Dr. Baniya's undergraduate Technical Communication and Documentation course. *30 attendees.*
13. **Webb, M.E.B.** and **Smith, A.** (2025, February). "Understanding Virginia Hospital Educators' Work Practices and Technology Needs." Invited talk with VT Institute for Creativity, Arts, and Technology. *30 attendees.*
14. **Webb, M.E.B.** (2025, April). "Poster Presentations for Research Conferences." Invited talk with the Virginia Tech College of Engineering Graduate and Professional Studies (COEGPS). *25 attendees.*
15. **Webb, M.E.B.** (2025, May). " Getting the Most Out of Engineering Conferences and Conference Resources." Invited talk with the Virginia Tech College of Engineering Graduate and Professional Studies (COEGPS). *25 attendees.*
16. Montilla, H., **Webb, M.E.B.,** Herron, C., Dimobi, C. (2026, April). "Shaping Your Career Path: Virginia Tech College of Engineering Graduate Alumni Panel" Invited talk with the Virginia Tech College of Engineering Graduate and Professional Studies (COEGPS). *45 attendees.*

Conference Presentations and Posters

1. **Webb, M.E.B.**, Schwarz, B., Sobti, S., King, P., Montalbano, B., Broadman, C., & Ghorbel, F. (2019, May). "BuoyBots: A Soft-Robotics Approach to Buoyancy Control." Presented at the Rice University Oshman Engineering Design Conference.
2. **Webb, M.E.B.,** Fraser, R., & Davies, H.R. (2022, March). "Psychosocial Factors in Disaster Research." Hosted a Roundtable Discussion at PRIMR Conference at the University of North Texas. *10 attendees.*
3. Fiore, S., Hartley, T., Granados, D., & **Webb, M.E.B.** (2022, August). Panel Session: Understanding and Addressing Interdisciplinary Learning: Perspectives from NSF Funded Research to Improve Team Science Education. Presented at the International Network for the Science of Team Science (SciTS) Conference [virtual].
4. **Webb, M.E.B.** (2022, October). "Facilitating Interdisciplinary Graduate Education in Disaster Resilience" NSF Research Traineeship Annual Meeting. *100 attendees.*
5. **Webb, M.E.B.** (2022, October). "Turning Convergence Research Topics into Convergence Research Questions and Teams" NSF Research Traineeship Annual Meeting. *70 attendees.*
6. Davies, H.R., **Webb, M.E.B.**, & Fraser, R. (2023, January). "Psychosocial Factors and Perishable Data Collection in Disasters: Creating Resilience through Research." Hosted a Roundtable Discussion at the Society for Social Work Research (SSWR) Conference. *10 attendees.*
7. **Webb, M.E.B.,** Scott, L., Salem, A., Baniya, S., & Powell, K. (2023). Towards Digital Justice: Developing US Citizenship Application & Website with Refugees & Immigrants Using Participatory Action Methods. Proceedings of the Conference on Community Writing Workshop, Denver, Colorado.
8. Smith, A.**, Webb, M.E.B.,** Rasberry, N., Baniya, S., & Ogbonnaya-Ogburu, I. F. (2025). Designing Technology to Support the Hospital Classroom. In Proceedings of the 2025 CHI conference on human factors in computing systems.

9. Baniya, S., **Webb, M.E.B.,** Scott, L., & Powell, K. (2026). Implementing Digital Justice to Support Refugee and Immigrant Communities. In Proceedings of the 2026 CHI conference on human factors in computing systems.
10. **Webb, M.E.B.,** Coso Strong, A, & Werth, A. (2026). From Framework to Field: Developing and Applying a Relational Agency Model for DBER Postdoc Professional Development. *In Proceedings of the 2026 Upstate NYDBER Conference.* Ithaca, NY.

Recognition

Honors and Awards

SEFI Early Career Researcher's Club: European Society of Engineering Education '25-'26

American Society of Engineering Education Graduate Studies Division (GSD) June 2025
Best Student Paper Award GSD, second year in a row
First author for "Tracking the Evolution of Interdisciplinary Identity-Based Motivation in Engineering Graduate Students: A Longitudinal Study" with co-author Dr. Marie C. Paretti
https://sites.asee.org/gsd/division-awards/

American Society of Engineering Education Graduate Studies Division (GSD) June 2024
Best Student Paper Award GSD
First author for "Understanding Ecosystems of Interdisciplinary Graduate Education through an Ecological Systems Approach" with co-author Dr. Marie C. Paretti

Virginia Tech
Nominated for Council of Graduate Studies (CGS) / ProQuest Distinguished Dissertation Award by Virginia Tech Graduate School ` 2026
VT Department of Engineering Bevlee Watford Outstanding Dissertation 2026
Nominated for VT Graduate School Outstanding Dissertation for Social Sciences 2026
Nominated for VT COE Outstanding Doctoral Student 2025
1st place VT COE Torgersen Outstanding Dissertation Awards May 2025
https://news.vt.edu/articles/2025/05/
https://torgersengradaward.vt.domains/2025-winners/
President's Honor Roll 2021-Present
Pratt Fellowship for Research Excellence, $2,000 2021-2022
Outstanding Capstone Design Project:
Bowman Sustainable Land Development CEE M.S. Program Spring 2023
https://lddi.cee.vt.edu/content/

Rice University 2015-2019
President's Honor Roll
Rice University Willy Revolution Award for Outstanding Innovation: Senior Capstone Award
https://oedk.rice.edu/Sys/PublicProfile/47555555/1063096
Oshman Engineering Design Engineering + Art Competition
https://oedk.rice.edu/news/7213404

Engineering Work Experience

Subsea Engineer, ExxonMobil 8/2019 – 3/2021

- Rotating Facilities Engineer to Norman Wells in NWT, Canada: Responsible for on-the-ground engineering activity and operations optimization
- Project Engineer Liza Phase 1 Subsea Pig Launcher: Developed, manufactured, and tested a $3M assembly that was deployed offshore Guyana for a remediation activity
- Subsea Hardware Lead Central South Dagi: Responsible for the design of all subsea hardware for a 14-well tieback offshore Sakhalin, Russia
- Internship Program Coordinator for Upstream Integrated Solutions: Planned and executed the internship program for 80 employees. Developed projects, assigned mentorship, and facilitated the development of new hires.
- Subject Matter Expert in Template Manifolds and Elastomer-Sealed Subsea Gaskets: Served as the main point of contact for technical questions and training on these two SME topics for all Exxon subsea projects.

Subsea Engineering Intern, ExxonMobil 5/2018 – 8/2018

- Designed an enhanced surveillance leak detection tool for potential methanol egress in subsea fields
- Performed a calculated risk assessment on critical path mid-water arches and recommended areas of focus for repair
- Oversaw the site receipt testing of production header isolation valves prior to assembly-level fabrication
- Scoped the 2018 cathodic protection survey for Mobil Equatorial Guinea using descriptive analytics in JMP

Process Engineering Intern, Air Liquide 5/2017 – 8/2017

- Designed an atmospheric pressure vessel that meets ASME safety requirements regarding natural gas disposal
- Collaborated with plant managers and process technicians to write new standard operating procedures for an off-site air separation facility

News

Cornell Chronical NY DBER Conference Overview: Conference aims to bridge divides in undergraduate education
https://news.cornell.edu/stories/2026/06/conference-aims-bridge-divides-undergraduate-education
https://as.cornell.edu/news/conference-aims-bridge-divides-undergraduate-education

Virginia Tech News: Class of 2025 Six students graduate with Ph.D. from Department of Engineering Education June 2025
https://news.vt.edu/articles/2025/06/Classof2025DepartmentofEngineeringEducation.html

Virginia Tech News Webb wins 1st place VT College of Engineering Torgersen Outstanding Dissertation Awards May 2025

https://news.vt.edu/articles/2025/05/eng-coe-torgersen-research-winners-2025.html
https://torgersengradaward.vt.domains/2025-winners/

Cornell University NSF-funded postdocs to research education across disciplines 2024

https://as.cornell.edu/news/nsf-funded-postdocs-research-education-across-disciplines

Virginia Tech Institute for Creativity, Arts, and Technology SEAD Grant Stories: Weaving Weeds, Exploring human-plant relationships through bio-fabrication

https://icat.vt.edu/projects/supported-research/sead-grants/2023-2024/student/weaving-weeds.html

Virginia Tech College of Liberal Arts and Human Sciences News Stories 2022

Internships help students understand the needs of refugees and migrants

Virginia Tech Bowman Sustainable Land Development Civil Engineering News 2023

Outstanding Civil Engineering Design Team: Bowman Sustainable Land Development Program Holds Annual Winter Meeting

Reuters: "Exxon exodus turns floating 'cube' into Internet meme" mention 2021

https://www.reuters.com/business/exxon-exodus-turns-floating-cube-into-internet-meme-2021-10-04/

Rice University Oshman Engineering Design Kitchen News Spring 2019

https://oedk.rice.edu/news/8792171: Engineering Design Showcase Willy Revolution Award
https://oedk.rice.edu/news/7213404: Engineering Art Competition

Fellowships

Colorado School of Mines Teaching with Heart Program 2024-2025

$500 fellowship to participate in experiential workshops focused on integrating empathy and care into STEM teaching

Institute for Creativity, Arts, and Technology Virginia Tech

Roger and Debbie West Student Research Grant, $1000 2023-2024
Weaving Weeds: Exploring Human-Plant Relations Through Bio-Fabrication

Society for Social Work Research

Doctoral Student Travel Award, $1500 2023

American Society of Engineering Education: Women in Engineering Division

Graduate Student Travel Grant, $1500 2023

National Science Foundation in Partnership with Virginia Tech

NSF Research Traineeship Fellowship Disaster Resilience 2022-2023

Virginia Tech

Travel Grant, Graduate and Professional Student Senate, $1500 2022-2023
Travel Grant, Center for Coastal Studies: Disaster Resilience 2022-2023

Rice University 2015-2019

Rice Engineering Alumni Association Buckley Sartwelle Scholarship, $12,000

Grant Writing Experience

NSF STEMEdIPRF Postdoctoral Fellowship Proposal **December 2024**

Transforming Graduate Education and Systems Perspectives on Interdisciplinary STEM Graduate Development and Programming

PI: Margaret Webb

- Led ideation and conceptual development
- Collaborated with mentors across universities
- Completed proposal writing and submission

NSF CAREER Award Grant Application work for Dr. Sweta Baniya, GRA **Summer 2024**

Theorizing Disaster Rhetorics: A Multi-Country Study for Disaster Response and Crisis Communication

PI: Dr. Sweta Baniya

- Conducted an extensive literature review to inform the theoretical framework and support the grant proposal.
- Collaborated on writing and editing key sections of the proposal, enhancing clarity, coherence, and alignment with NSF priorities.

Website Development Experience

Personal Research Website: https://webbmargaret.github.io/

- My academic portfolio site, with pages for Research Overview, Publications, Talks, Teaching, and CV, plus links to your Google Scholar, ORCID, ResearchGate, and LinkedIn profiles.
- **Built with:** Jekyll (a static site generator) using the **AcademicPages** theme, a popular fork of the Minimal Mistakes Jekyll theme built specifically for academic personal sites. It's hosted free on **GitHub Pages**, which auto-builds the Jekyll site from my repo on every push.

Upstate NYDBER Conference Website: https://nydber2026.github.io/nydberconference2026/

- An event site for the Upstate NY DBER Conference (June 11–12, 2026, Cornell), with Home, Program, Directory, Posters, and Gallery pages. It covers the conference theme, NSF funding, and a downloadable flyer/logo assets.
- **Built with:** Static HTML/CSS, also hosted on **GitHub Pages** (org-level repo `nydber2026`).

DBER Network Analysis Website: https://dber-scholars-network/index.html

- An interactive, static visualization of a DBER scholars dataset — a **D3.js force-directed "hub-and-spoke" network graph** grouping scholars by institution, program, DBER field, and PhD era. It includes an About page explaining the dataset/limitations and a Contribute page (mailto-based form for people to add themselves or request corrections).
- **Built with:** Python (`build_data.py`) to process a source CSV into JSON (`scholars.json`, `groups.json`), then a static HTML/CSS/**JavaScript (D3.js)** front end.

DBER Job Webscrape Website: https://dber_postdoc_job_market_report.html

- A data report analyzing 15+ years of academic job postings, built by scraping and classifying listings from two community-run job boards: the **PER Jobs blog** (Physics Education Research) and the **Engineering Education job wiki**.
- **Built with:** A **Python web-scraping pipeline** pulling postings from the two job boards, likely parsed/cleaned and classified (by role type, institution, subfield, etc.) into a dataset, then rendered as a static HTML report — hosted on **GitHub Pages** like my other sites.

Service

Service Roles

American Society of Engineering Education Graduate Studies Division Secretary 06/2025 - Present

Rice University Engineering Alumni Association 05/2025 - Present
Education Research Consultant

Virginia Tech Engineering Outstanding Doctoral Student Award Committee Spring 2024

VT Engineering Dean's Search Committee for Assistant Director Grad Studies Fall 2024

Virginia Tech Graduate and Professional Student Senate (GPSS) 2023-2024
Representative for the Department of Engineering Education
Voting Member of the Commission on Faculty Affairs

American Society of Engineering Education Student Division Best Paper Chair 2023-2024

Virginia Tech Chapter of American Society of Engineering Education (ASEE@VT)
Vice President 2022-2023

Natural Hazards Engineering Research Infrastructure Graduate Student Council (GSC)
Chair of DEI and President of Social Science Research Group 2022 - 2024

ExxonMobil, Subsea Engineering Department Houston, TX
United Way Champion 2019-2020

Children's Assessment Center (CAC) Houston, TX
Volunteer *Program Coordinator* 8/2018-5/2020

Rice University, Department of Mechanical Engineering Houston, TX
Co-Founder of Mechanical Engineering Undergraduate Advisory Board 2016/2019
Student Representative to the Credit Cap Committee 8/2015 – 5/2016
Student Ambassador to Rice Engineering 2015 – 5/2019
Peer Career and Peer Academic Advisor 6/2015-5/2019

Professional Membership

VT Community of Practice for Professional Development 12/2024 – 5/2025
AAAS/Science Program for Excellence in Science 10/2024 - Present
American Society for Engineering Education 8/2021 – Present
Graduate Studies Division
Multidisciplinary Engineering Division
Student Division

Natural Hazards Research Infrastructure Graduate Student Council 8/2021 – 5/2024
DEI Committee
Social Science Research Committee
Rice University Engineering Alumni Association 05/2019 - Present
Society of Petroleum Engineers 8/2021 – 8/2022
ExxonMobil ToastMasters 8/2019 – 12/2020
ExxonMobil Black Employees Success Team 8/2019 – 12/2020
ExxonMobil PRIDE 8/2019 – 12/2020
ExxonMobil Women's Interest Network 8/2019 – 12/2020
Society of Women Engineers 8/2015 – 8/2019
American Society of Mechanical Engineers 8/2015 – 8/2019
Rice Creative Society 8/2015 – 5/2019

Professional Development

Conference Attendance, Paper Reviews, Moderation

Collaborative Network for Engineering and Computing Diversity Conference 2022
New Orleans, LA
In-Person Attendee

The Natural Hazards Workshop Spring 2022
Boulder, CO at UC Boulder CONVERGE Facility
Virtual Attendee

Frontiers in Education (FIE) Conference, Uppsala, Sweden Fall 2022
In-Person Attendee
Reviewer

Australasian Association for Engineering Education (AAEE) Conference
Virtual Attendee Winter 2021 and 2022
Reviewer Winter 2022
PhD Research (HDR) Symposium 12/2021

American Society of Engineering Education (ASEE Conference)
In-Person Attendee Summer 2022 - Present
Reviewer
Moderator

Natural Hazards Research Infrastructure Graduate Student Council Conference 2023
Virtual Attendee

Cornell University Systems Engineering Education Summit 2025
In-Person Attendee and Moderator

Professional Development Workshops Attended

Virginia Tech Newman Library — Virtual
Evidence Synthesis Reviews 101 — 9/2021

Virginia Tech Dissertation Institute Summer 2024 — In-Person

Colorado School of Mines Teaching with Heart Program — Virtual 2024-2025